

4.4.2 Incumbent Advantage versus Market Entry Barriers

Electric vehicle deployment benefits established automotive manufacturers, existing utilities generating revenue regardless of generation mix, and natural gas producers serving marginal generation. These actors possess lobbying capacity and political relationships to shape subsidy structures.

Biomass-hydrogen pathways would benefit potential new entrants: rural feedstock aggregators, small-scale gasification operators, distributed refueling station developers. These actors lack comparable political influence, creating systematic bias in allocation decisions absent explicit countermeasures (parliament-based deliberation, performance-based subsidy criteria).

4.4.3 Cost Competitiveness versus Systemic Benefits

Direct cost comparison shows biomass-hydrogen 3–5× more expensive than fossil baselines absent carbon pricing. However, advantages in emission profile certainty (not grid-dependent), distributed deployment enabling rural participation, domestic feedstock security, and network resilience represent systemic benefits not captured in \$/kg metrics.

The policy choice is whether these systemic benefits justify public investment to close cost gaps. Current structures privilege technologies with lower direct costs but higher systemic risks (grid dependency, benefit concentration, incumbent entrenchment). Alternative structures emphasizing lifecycle performance and benefit distribution would favor pathways with higher direct costs but superior systemic characteristics.

5 Conclusion: Toward Equitable Decarbonization Architecture

This analysis establishes three core findings that jointly characterize the climate investment allocation challenge:

Finding 1: Benefit Concentration as Structural Feature. Clean energy investment systematically concentrates benefits among incumbent actors and higher-income households through program design features rather than technology characteristics. The top income quintile captured 60% of \$47 billion in clean energy tax credits since 2006 despite nominal equity objectives. This pattern repeats across infrastructure categories (broadband, education, economic opportunity), suggesting common structural causes: program requirements privilege populations possessing tax liability, home ownership, upfront capital access, and institutional sophistication.

Finding 2: Categorical Framing Enables Evasion. “Zero emission vehicle” classification creates coordination equilibrium where manufacturers avoid deploying available cleaner production pathways (renewable-powered batteries, green steel, responsible mining reducing embodied emissions by 11 tonnes CO₂e per vehicle—92% of manufacturing footprint), policymakers claim climate progress without mandating lifecycle accountability, and citizens experience satisfaction from consumer transactions rather than sustained civic engagement. This three-actor evasion dynamic produces symbolic achievement while systematically preventing emission reductions within technical and economic reach.

Finding 3: Alternative Pathways Exist But Face Systematic Disadvantages. Biomass-to-hydrogen gasification achieves competitive levelized costs (\$9.57–10.87/kg at 100-unit cluster scale) approaching fossil+CCS ranges with moderate carbon pricing. Pathways exhibit superior characteristics: lifecycle emissions independent of grid composition (0.5–2.5 kg CO₂e/kg H₂ for residue-based feedstock), distributed deployment model enabling rural participation, domestic feedstock reducing geopolitical exposure. However, incumbent advantages in policy access,